Gait Variability and Fall Risk in Community-Living Older Adults: A 1-Year Prospective Study

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ABSTRACT. Hausdorff JM, Rios DA, Edelberg HK. Gait variability and fall risk in community-living older adults: a 1-year prospective study. Arch Phys Med Rehabil 2001;82:1050-6.

Objective: To test the hypothesis that increased gait variability predicts falls among community-living older adults attending an outpatient clinic.

Design: Prospective, cohort study.

Setting: Three outpatient geriatric clinics.

Participants: Fifty-two community-living, ambulatory men and women aged $\geq$ 70 years.

Interventions: Not applicable.

Main Outcome Measures: Subjects walked at a normal pace for up to 6 minutes wearing force-sensitive insoles that measured the gait rhythm on a stride-to-stride basis. Afterward, subjects reported fall status on a weekly basis for 1 year. The primary outcomes were the association between measures of the stride-to-stride fluctuations in gait rhythm and (1) subsequent falls during a 12-month follow-up period and (2) potential contributing factors.

Results: Almost 40% of the subjects reported falling during the 12-month follow-up period. Stride time variability was 106 $\pm$ 30ms in subjects who subsequently fell ($n = 20$) and 49 $\pm$ 4ms in those who did not experience a fall ($n = 32$) during the 12-month follow-up period ($p < .04$). Logistic regression also showed that stride time variability predicted falls ($p < .05$). Stride time variability correlated significantly with multiple factors including strength, balance, gait speed, functional status, and even mental health, but these other measures did not discriminate future fallers from nonfallers.

Conclusions: These findings show both the feasibility of obtaining stride-to-stride measures of gait timing in the ambulatory setting and the potential use of gait variability measures in augmenting the prospective evaluation of fall risk in community-living older adults.

Key Words: Accidental falls; Aged, 80 and over; Locomotion; Rehabilitation.

DESPITE EXTENSIVE PREVENTIVE efforts, falls continue to be a major source of morbidity and mortality among older adults.[1,2] Falls often lead to serious injuries such as hip fractures, hospitalization, and death.[1-3] Even when no serious injury occurs, the resultant fear of falling and self-imposed restrictions in mobility and function may be contributing factors for nursing home admission.[4,5] In light of the significant social and economic ramifications of this important public health problem,[5,6] interventions aimed at reducing fall risk are being proposed and clinically tested, with varying degrees of success.[7-12] The wide-ranging effects of falls and the potentially high cost of these interventions necessitate identification and targeting of those persons at greatest risk of falling.[2] Some measures of balance and physical function appear to be useful in prospectively identifying potential fallers,[13-15] but there remains much room for improvement of targeting efficiency so that health care resources are allocated to the most vulnerable individuals.

Falls among older adults often occur during walking, and gait dysfunction is included among the many risk factors for falls.[16,17] Although marked differences in the gait patterns of fallers and nonfallers might be anticipated, few quantitative studies have found measures of gait that distinguish between these 2 groups. Moreover, few quantitative measures of gait have been able to identify prospectively those older adults who are at greatest risk of falling. For example, despite a recent finding that peak joint torques apparently increase in older adults with a history of falls,[18] other kinematic and kinetic investigations have found no differences between fallers and nonfallers and no predictive value for parameters that describe joint kinematics, kinetics, or gait cycle timing.[19,20]

These quantitative studies of walking in elderly fallers have largely focused on properties of an average stride or gait cycle as measured over a few cycles. However, many features of walking actually fluctuate from 1 stride to the next, even under healthy conditions.[21-23] Among fallers, these stride-to-stride fluctuations may be characteristically increased,[24,25] suggesting a possible role for quantitative measures of gait variability in fall risk assessment. Indeed, we previously found increased gait variability among community-living older adults with a history of falls.[26] Maki[27] observed that although gait speed and other parameters describing properties of the "average" stride (eg, the duration of average gait cycle, the mean stride time) were associated with a fear of falling, markers of stride-to-stride variability were apparently predictive of future falls among older residents of an assisted-living facility.

The present study was designed to extend these studies of gait variability. Even though it appears that gait variability is increased among certain elderly fallers, a systematic study of the factors that contribute to this variability is lacking. Moreover, it is unclear if measures of gait variability could be used in a clinical setting to help to prospectively identify potential fallers. The primary objectives of this investigation were, therefore, (1) to test the hypothesis that increased gait variability in community-living older adults attending an outpatient geriatric assessment clinic predicts falls, and (2) to examine the correlates of gait variability in this population. We also sought to

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begin comparing the predictive value of gait variability measures with other measures of fall risk.

METHODS

Subjects

This prospective cohort study was conducted at 3 hospital-affiliated geriatric clinics in the greater Boston area. Patients were recruited from the Beth Israel Deaconess Gerontology Group and the Youville Hospital and Rehabilitation Center outpatient clinics. All ambulatory, community-dwelling patients age 70 years and older who presented to 1 of the study sites were eligible for the study. Patients were excluded if they had severe cognitive impairment (unable to follow simple directions), were nursing home residents, had a life expectancy of less than 1 year, or refused to participate. All subjects provided informed, written consent. The institutional review boards of the Beth Israel Deaconess Medical Center and Youville Hospital and Rehabilitation Center approved the study.

Procedures

After informed consent was obtained, subjects were evaluated, gait was tested, and chart review was performed. After this assessment, the research staff contacted each subject weekly for 52 weeks to obtain information regarding any falls[28] or related incidents.

The evaluation included measurement of gait variability. To characterize more fully the functional and neurophysiologic status of the subject population, factors that might contribute to gait variability, as described below, and factors that have been associated with fall risk were also included in the evaluation. These included demographic data (age, gender, height, weight, level of formal education) as well as health status, mental status, self-rated quality of life (QOL), functional status, muscle strength, balance, and gait. The Charlson Comorbidity Index (CCI),[29] body mass index (BMI), and number of prescription medications were used as markers of health status. Mental health and emotional well-being were assessed with the Mini-Mental Status Examination (MMSE),[30] and the Geriatric Depression Scale (GDS),[31] respectively. The Medical Outcomes Study Short-Form Health Survey (SF-36)[32] was used to evaluate 8 domains of perceived health-related QOL: physical functioning, role–physical, bodily pain, general health, vitality, social functioning, role–emotional, and mental health. Physical activity levels were estimated from the Physical Activity Scale for the Elderly (PASE).[33] Self-report of the ability to perform activities of daily living (ADLs) and instrumental activities of daily living (IADLs) were assessed by using the Katz ADL[34] and Lawton IADL[35] scales. The Short Physical Performance Battery (SPPB)[36] was used as a performance-based measure of function. Subjects were also asked if they had a fear of falling (yes, no), and attitudes toward falls were further assessed by using the Falls Efficacy Scale (FES),[37] a continuous measure that describes how confident one is in performing different ADLs without falling. Grip strength was measured by using a hand-held dynamometer[a] and, in a subset of patients (the last 28 studied), isometric knee extension strength was also tested by using another hand-held dynamometer.[b] To adjust for physical differences, strength measures were normalized to body weight.[27] Balance and gait were studied with a variety of static and dynamic measures: the Functional Reach Test,[38] 1-legged stance time (eyes open and closed), tandem stance time, the Timed Up and Go,[39] and the Performance-Oriented Mobility Scale.[40]

Assessment of Gait Variability

The protocol and methods were similar to those used previously.[26,41] Briefly, subjects were instructed to walk at their normal pace on level ground for as long as possible for up to 6 minutes. To measure the gait rhythm and the timing of the gait cycle, force-sensitive insoles[c] were placed in the subject's shoe.[42] These inserts produce a measure of the force applied to the ground during ambulation. A small, light-weight ($5.5 \times 2 \times 9$cm; 0.1kg) recorder was worn on the ankle and held in place by using an ankle wallet. An on-board analog-to-digital converter (12 bit) sampled the output of the footswitches at 300Hz and stored the data. Subsequently, the digitized data were transferred to a UNIX workstation for analysis by using software that extracts the initial contact time of each stride.[42] With this information, the stride time or duration of the gait cycle (time from initial contact of 1 foot to subsequent contact of same foot) and swing time were determined for each stride. Each subject's mean gait speed was determined by dividing the total distance walked by the duration of the walk time. The stride and swing time histories (time series) were analyzed to assess quantitatively different aspects of the stride time dynamics.[41] All of the gait variability analyses were performed in a blinded fashion without knowledge of any other subject information.

To study the intrinsic dynamics of the gait rhythm, the time series of the stride and swing time were analyzed as follows.[26] The first 10 seconds of the data were excluded to minimize start-up effects and a median filter was applied to remove data points that were 3 standard deviations (SDs) higher than or lower than the median value.[26] Each subject assessment typically included measurement of several hundred strides. The average stride and swing time were determined for each subject and the magnitude of the variability was calculated as follows. Stride time variability and swing time variability are defined here as the SD of each subject's stride time and swing time time series, respectively. These measures assess the magnitude of the deviations of the stride and swing time with respect to each subject's mean value. Note that these measures are, by definition, normalized with respect to the length of the time series (the number of strides) so that these measures are independent of the duration of the subject's walk. Another common measure of the magnitude of the variability is the coefficient of variation (CV). Here, we report the results by using the SD because for both stride time and swing time, the SD and the CV of the time series correlated highly with one another ($r > .95$, $p < .001$) and all results were similar with both measures.

A complimentary measure of the stride-to-stride variability—the inconsistency of the variance of the stride time—was used to evaluate how the variability changes with time, independent of the fluctuation magnitude. To minimize effects of subject-to-subject differences in data length, mean, or variance, the first 100 points of each subject's time series were normalized with respect to the mean and SD, yielding a new time series with the mean equal to 0 and the SD equal to 1, but with different dynamic properties depending on how the local property changed with time. More specifically, the mean of each time series was subtracted from each value in the time series (so that the new time series has a mean of 0) and the time series was divided by its SD (so that the SD of the new time series was, by definition, equal to 1). This normalized time series was then subdivided into 20 segments and in each segment the (local) SD was computed. The inconsistency of the variance, defined as the SD of these 20 SDs, was then calculated to estimate the dispersion of the local variance. Thus, this metric provides a measure of the inconsistency of the magnitude of the fluctuations, independent of the overall variance (the fluctuation magnitude) of the original time series. Higher values indicate more inconsistent changes with time; the local variance is fluctuating, even after adjusting for the mean and SD of the original time series.

Statistical Analysis

For continuous data, the Wilcoxon rank test was used to test for statistical differences between subjects who experienced a fall and those who did not. Correlation between variables was tested by using Spearman's correlation coefficient. As suggested by Maki,[27] nonparametric tests were used because the results were not likely to be normally distributed and these tests make no assumptions about the underlying distribution of the data. (Similar results were obtained by using parametric tests both before and after log transformation.) Following convention,[27] logistic regression was used to test the predictive value of those measures that differed significantly between fallers and nonfallers. Survival analysis was performed to study the relationship between gait variability and the time to first fall by using Cox's partial likelihood method. A 2-sided p value of .05 or less was considered statistically significant. Unless otherwise specified, group results are summarized as mean $\pm$ SD. Statistical analysis was performed by using SAS software, version 7.0.[d]

RESULTS

Subject Characteristics

Fifty-two subjects (36 women, 16 men) were tested and then followed-up for 12 months. There were no deaths and all subjects were in contact with the research staff on a weekly basis throughout the 1-year follow-up period. The mean age of the subjects was 80.3 $\pm$ 5.9 years. Mean height and weight were 161 $\pm$ 11cm and 66.9 $\pm$ 16.6kg, respectively. In general, the subjects were relatively healthy (CCI score, 0.9 $\pm$ 1.3), cognitively intact (MMSE score, 27.8 $\pm$ 4.1), and able to perform both basic and instrumental ADLs (Katz ADL score, 6.3 $\pm$ 1.0; Lawton IADL score, 21.3 $\pm$ 3.9). Lower extremity function was also relatively intact; the mean gait speed was .84 $\pm$ .29m/s (gait speed in a group of similarly aged healthy adults free of musculoskeletal, neurologic, or cardiac disease has been reported[18] as .82 $\pm$ .02m/s).

During the 52-week follow-up period, 20 subjects reported falling 1 or more times. Among the first falls reported by each subject, 15 of the 20 falls (75%) occurred while the subject was walking and 4 (20%) occurred during other activities in which the center of gravity was displaced from the base of support (eg, turning, bending, stretching). Fourteen of these 20 falls (70%) took place in the afternoon, 5 (25%) took place in the morning, and 1 (5%) took place in the evening. Nine of the 20 subjects who fell reported falling only once and 11 fell multiple times during the follow-up period.

Gait Variability and Fall Risk

Measures of gait variability were significantly increased in those who subsequently fell ($n = 20$) compared with those who did not ($n = 32$), as summarized in figure 1. An example of the increased stride-to-stride fluctuations observed in a subject who fell during the follow-up period is shown in figure 2. (Note that stride time variability was twice as great in subjects who experienced multiple falls compared with those who reported 1 fall, but the inconsistency of the variance was similar in both of these groups of fallers.) Logistic regression analysis showed that measures of gait variability were also predictive of future falls (table 1). Survival analysis indicated that subjects with increased gait variability were likely to fall sooner than those subjects who had less variability in gait during their clinical assessment (table 2).

Although measures of gait variability were different in those who fell, the 2 groups did not differ significantly with respect to age, gender, height, weight, BMI, health status (ie, CCI, number of medications), mental health (MMSE, GDS), level of education, physical activity levels, or ability to perform ADLs or IADLs. Gait parameters based on the typical average value were also similar in those who did and did not fall during the 1-year follow-up period. For example, the duration of the walk during the gait assessment was very similar in fallers and nonfallers ($p = .52$). Although there were some trends in the data (table 3), neither mean stride time, swing time, percentage swing time, and gait speed, nor measures of balance, lower extremity function (eg, Functional Reach Test, Timed Up and Go, Performance-Oriented Mobility Scale), and physical performance (SPPB) were significantly different in the fallers compared with the nonfallers.

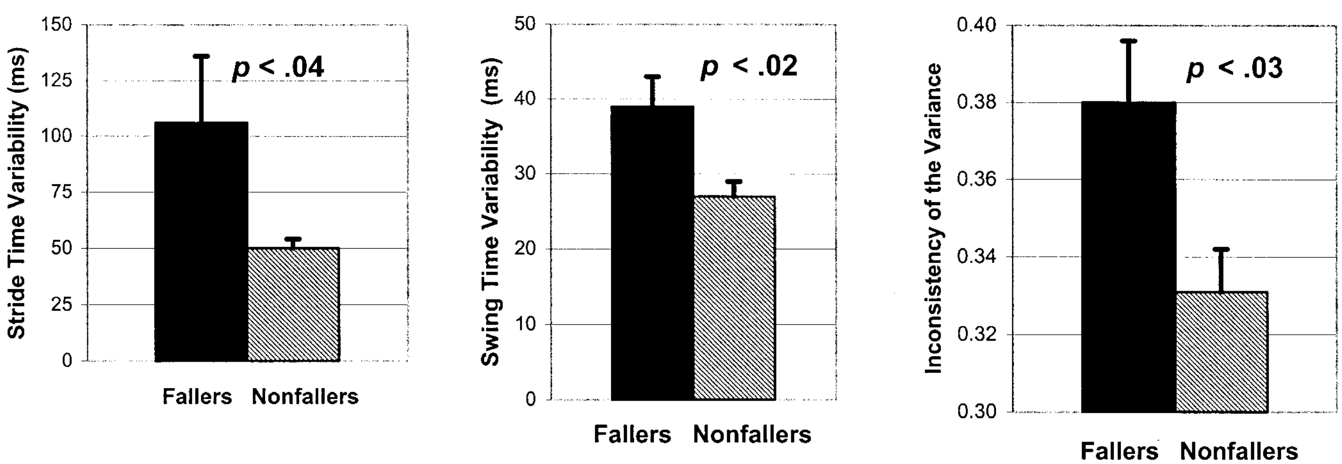

Fig 1. Measures of gait variability in subjects who fell and those who did not during the 12-month follow-up period. For all 3 measures, variability was increased significantly in those subjects who subsequently experienced a fall. Shown are the group means. The error bars reflect the standard error of the means.

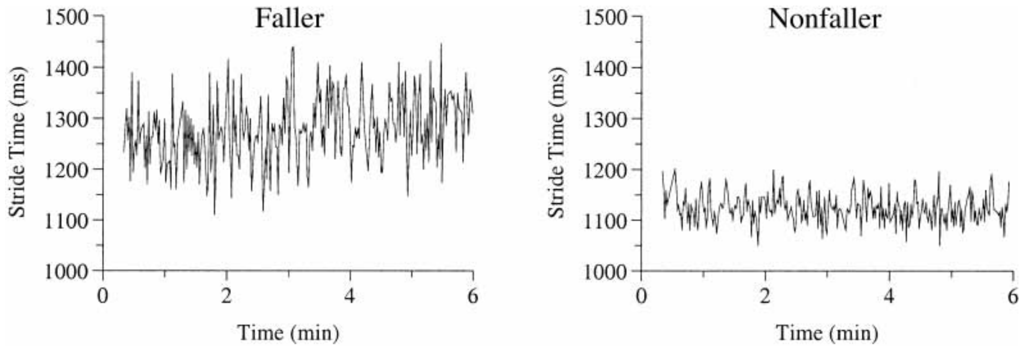

Fig 2. Stride-to-stride fluctuations in stride time as measured in a 78-year-old man who fell during the 12-month follow-up period (stride time variability = 66ms, inconsistency of the variance = .39) and an 84-year-old man who did not fall (stride time variability = 29ms; inconsistency of the variance = .28).

Correlates of Gait Variability

Stride time variability was associated with many factors; in general, increased variability correlated with poor health status and decreased performance (table 4). Stride time variability was related to health status and mental status. Stride time variability tended to increase with age and was lower with increased strength: grip strength and knee extension strength. Stride time variability was related to balance, gait, lower-extremity function, physical activity level, and self-report and performance-based measures of functional status. Stride time variability was also inversely related to 5 of the 8 health-related QOL measures: physical function, role–physical, general health, social function, and vitality. Stride time variability was not different in those who reported a fear of falling, but it was associated with confidence in one's ability to perform activities without falling. Stride time variability did not differ between men and women.

Similar results were obtained for swing time variability. Indeed, stride time variability and swing time variability correlated highly with one another ($r = .89$, $p < .0001$). These measures did not, however, correlate with the inconsistency of the variance ($p > .05$). Instead, the factors associated with stride time variability and swing time variability were generally different from those associated with the inconsistency of the variance.

As was the case with stride time variability, an increased inconsistency of the variance was associated with poor health and functional status. The inconsistency of the variance was marginally related to depressive symptoms ($r = .27$, $p = .053$), self-report of vitality ($r = -.27$, $p = .06$), Performance-Oriented Mobility ($r = -.24$, $p = .07$), and Timed Up and Go ($r = .26$, $p = .053$). The inconsistency of the variance was significantly related to self-report of social functioning ($r = -.34$, $p = .018$) and balance, specifically, eyes-closed 1-legged stance time ($r = -.30$, $p = .028$). The inconsistency of the variance was not different in men or women and was not related to fear of falling or fall efficacy.

DISCUSSION

This quantitative study of gait variability has several key findings. First, among community-living older adults attending an outpatient geriatric clinic, increased gait variability is associated with an increased risk of future falls. In fact, the likelihood of falling was increased about fivefold with only a moderate increase in stride time variability (table 1). Second, in this population of older adults, measures of variability were not only associated with many factors that are intuitively related to fall risk, such as strength, balance, and gait, but also with vitality and mental status. Third, stride time variability was also strongly associated with self-report and performance-based measures of functional status, some of which have been shown to predict clinical outcomes such as morbidity and nursing home admission.[43]

To some extent, the present findings are consistent with previous results obtained in slightly different settings and populations. Feltner et al[20] and Maki[27] found that gait speed was not predictive of future falls and Maki observed a similar association between variability measures and risk of future falls. In the Maki[27] study, older residents of an assisted-living facility were tested in a gait laboratory. This study extends these findings in several ways. Although a categoric measure of fear of falling (yes/no) was unrelated to gait variability, a continuous self-efficacy measure of one's ability to perform ADLs without falling was associated with stride time variability. This suggests that stride-to-stride variability may not be completely independent of neuropsychologic function and fear of falling. Indeed, clinical depression has been associated with fall risk, independent of medication usage,[44] and there appears to be a correlation between vitality and gait variability. Furthermore, the present study shows the feasibility of obtaining measures of gait variability in a clinical setting and shows the relationship between gait variability and future falls in community-living older adults.

The finding of a relationship between gait variability and fall risk in this clinical setting is somewhat surprising. In geriatric medicine clinics, patients are typically assessed by a clinical team that is acutely aware of the problems associated with falls

Table 1: Predictive Value of Gait Variability Measures

	Odds Ratio*	95% Wald Confidence Intervals	p†
Stride time variability	5.3	1.01–27.2	.04
Swing time variability	2.2	1.1–4.4	.02
Inconsistency of the variance	2.1	1.1–4.1	.02

* Odds ratio of falling based on a 1 SD change in variability measure.
† p value based on logistic regression.

Table 2: Gait Variability Measures Predict Time to Fall

	Wald Chi-Square	p*
Stride time variability	9.5	.002
Swing time variability	7.5	.006
Inconsistency of the variance	6.0	.001

* p value based on survival analysis by using proportional hazards model.

Table 3: Balance and Gait in Subjects Who Did and Did Not Fall During the 1-Year Follow-Up Period*

	Nonfallers	Fallers	*p*
Mean gait speed (m/s)	.91 ± .24	.71 ± .33	.073
Performance-oriented mobility	25.6 ± 3.7	22.5 ± 6.9	.088
Functional reach (in)	11.5 ± 4.2	10.1 ± 6.5	.108
Timed Up and Go (s)	14.7 ± 3.0	21.9 ± 14.3	.130
SPPB	7.0 ± 2.3	5.9 ± 3.1	.167

* In contrast to the results shown in figure 1, these measures of gait were not different in fallers and nonfallers.

in older adults and knowledgeable about the various ways in which fall risk can be reduced. All of the physicians who cared for the patients in our study were board-certified geriatricians who had received extensive training in clinical geriatrics. Nevertheless, almost 40% of the study subjects reported falling during the 1-year follow-up period. This may be a result of selection bias because patients seen for primary care or consultation in a geriatric clinic may be more mentally or physically frail. It may also reflect both the challenges of reducing fall risk and the inherent difficulty of identifying older persons most likely to benefit from an intervention. Thus, the association between gait variability and fall risk in the present population underscores the potential use of quantitative measures of gait variability in fall risk assessment.

Increased stride time variability was associated with many other measures of fall risk, physical performance, and functional status. Although some of these other measures tended to be different in fallers and nonfallers, they did not distinguish the 2 groups in the sample population. Of note, stride time variability was also related to neuropsychologic function: vitality, mental status, and social function. Because it relies on central and peripheral inputs and feedback, as well as neuropsychologic function, the stride time can be viewed as a final, integrated output of the locomotor system. Gait variability measures may therefore provide a sensitive "assay" of neurodynamics, perhaps revealing increased variability (instability) even when the component systems that contribute to gait variability only show more subtle changes. This dependence on a host of physiologic and neuropsychologic factors may help to account for the observed predictive value of gait variability measures.

The present study has several limitations. The relatively small sample size may have resulted in a type II error and a failure to detect other differences between future fallers and nonfallers. Indeed, because other indices of fall risk are sensitive to many of the same components of balance and gait as those of gait variability, these measures might have been expected to differ significantly between the fallers and nonfallers. Gait variability may be more directly related to fall risk whereas other indices are less specific, measuring factors that do not necessarily increase the risk of falls. A larger sample size would minimize the possibility of type II errors, and allow further study of (1) the relationship between gait variability, other indices of fall risk, and the factors that contribute to falls; and (2) the specificity and sensitivity of different predictors of fall risk.

Table 4: Correlates of Stride Time Variability

		r	*p*
Health status	CCI	.47	.0005
	No. of medications	.50	.0002
Mental status	MMSE	−.47	.0005
	Age	.45	.0005
Strength	Grip strength	−.40	.003
	Knee strength	−.38	.04
Balance	Eyes open, 1-legged stance time	−.62	.0001
	Eyes closed, 1-legged stance time	−.51	.0001
	Tandem stance time	−.49	.0002
	Functional reach	−.35	.01
Lower-extremity function	Gait speed	−.82	.0001
	Stride time	.86	.0001
	Up and Go time	.78	.0001
	Performance-oriented mobility	−.75	.0001
	SPPB	−.74	.0001
Health-related QOL	Physical–function	−.53	.0003
	Role–physical	−.39	.006
	General health	−.41	.005
	Social function	−.36	.011
	Vitality	−.41	.003
Functional status	ADLs	.35	.0001
	IADLs	−.53	.0001
	Physical activity level (PASE)	−.43	.003
	FES	−.56	.0001

NOTE. Those variables that were significantly related to the stride time variability are listed. The *r* value is the Spearman correlation coefficient.

In addition, a larger sample size might have enabled us to assess the generalizability of these findings and to investigate individual differences between fallers. The relationship between increased gait variability and fall risk may be a function of the specific circumstances and pathophysiology that result in a fall.[27] Previous work has established a link between extrinsic or environmental factors (eg, a loose rug) and fall risk.[24] Future studies should examine the relationship between extrinsic factors, gait variability, and fall risk. Extrinsic factors that might otherwise pose little challenge may precipitate a fall in individuals with increased gait variability. Prospective follow-up of a cohort of older adults over an extended period of time (eg, years) would show how changes in gait variability and fall risk vary over time and with the aging process. Although difficult to perform in a clinical setting, concomitant study of kinematics and kinetics may also help to elucidate further the mechanisms that contribute to gait variability and fall risk. The present study was not designed to determine whether increased gait variability increases the risk of falling or simply serves as a marker of underlying deficits that predispose to falls. Differences between the factors that contribute to stride time variability and the inconsistency of the variance also warrant additional investigation. Future study of a larger sample population might help to elucidate the association between fall risk and gait variability and the potential clinical use of measures of the stride-to-stride fluctuations in gait.

CONCLUSION

It has been suggested that increased gait variability is not an inevitable part of the aging process.[21,45] The present findings support the idea that changes in gait variability are likely to reflect underlying disease processes rather than age-related changes. Previous studies have shown that exercise may restore physiologic capacity and reduce gait variability[46] and that exercise, Tai Chi, and other interventions may also be effective in minimizing fall risk.[7,8,10,11,47] Although further research is needed to understand more completely the wide variety of factors that contribute to gait variability and its relationship to fall risk, our study suggests that measures of the stride-to-stride fluctuations in gait may be used to help to augment the identification of older adults most likely to benefit from these promising interventions.

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Suppliers

a. Jamar model 5030J1; Sammons Preston, AbilityOne Corp, 4 Sammons Ct, Bolingbrook, IL 60440.
b. Chattilon model MSE-100; John Chattilon and Sons Inc, 7609 Business Park Dr, Greensboro, NC 27409.
c. Interlink Electronics, 546 Flynn Rd, Camarillo, CA 93012.
d. SAS Institute Inc, SAS Campus Dr, Cary, NC 27513-2414.